The `energese` Package User Guide

1 Introduction

`energese` renders H.T. Odum's Energy Systems Language (ESL) diagrams from a description of the system rather than from drawing commands. You state what the components are and how energy moves between them; the package places them and draws the result.

A system is described by three things: *nodes* (the components), *edges* (the pathways between them) and *metadata* (diagram-wide settings). That description can be written as JSON or directly in TEX. Both are parsed into the same model and rendered by the same code, so the two are interchangeable and produce identical output.

2 Requirements

LuaLATEX (TEX Live 2023 or later), TikZ with the `positioning`, `arrows.meta`, `shapes.geometric`, `calc` and `backgrounds` libraries, and the `luacode`, `environ` and `fontspec` packages. The `dkjson` Lua library ships with the package.

Put `energese.sty`, `energese-core.lua`, `energese-shapes.tex` and `dkjson.lua` somewhere on the TEX search path — typically the directory holding your document.

3 Authoring a diagram

There are three ways in. The first two share the layout engine; the third bypasses it.

3.1 From a JSON file

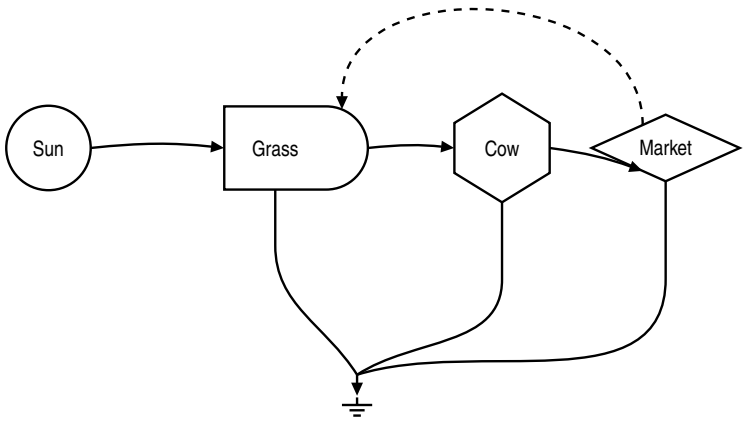
```
\renderEnergese{examples/simple_system.json}
```

An optional argument is passed straight through to the underlying `tikzpicture`:

```
\renderEnergese[scale=0.8]{examples/simple_system.json}
```

A `scale` shrinks the diagram as one picture, labels included: the layout is computed for symbols at their full size, so they must scale with the coordinates, and `transform shape` is added for you. A `xscale` or `yscale` is refused, because it would distort the symbols. To space a diagram out rather than shrink it, change `column_spacing` and `row_spacing`; to resize the symbols, `\energeseunit` (Section 10).

The path is resolved relative to the current directory first, then through `kpsewhich`, so a document is not tied to the directory it is compiled from. That call produces:



3.2 From inline JSON

For a one-off diagram that does not warrant its own file:

```
\begin{energeseData}
{
  "nodes": [
    {"id": "sun",   "type": "source",   "Tr": 1,   "label": "Sun"},
    {"id": "grass", "type": "producer", "Tr": 100, "label": "Grass"}
  ],
  "edges": [
    {"from": "sun", "to": "grass", "type": "energy"}
  ]
}
\end{energeseData}
```

3.3 From TEX

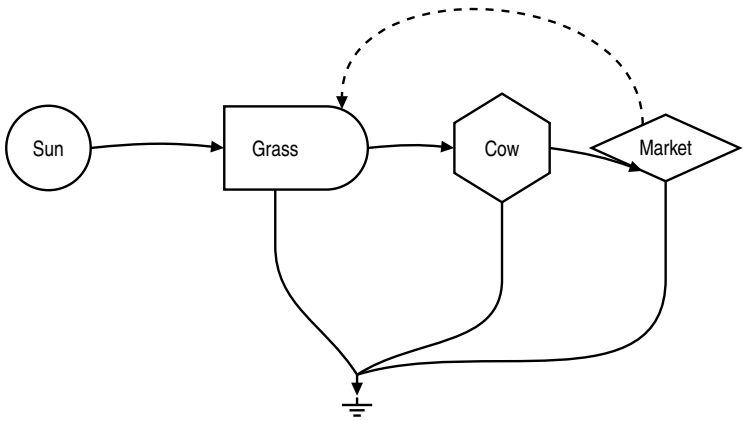
The `energese` environment describes the same model with declarative macros. Keys are exactly the JSON field names, so anything expressible in one front end is expressible in the other.

```
\begin{energese}
    \esmeta{column_spacing=3.0, row_spacing=1.5, show_heat_sink=true}

    \esnode{sun}{source}         {Tr=1,    label={Sun}}
    \esnode{grass}{producer}     {Tr=100,  label={Grass}}
    \esnode{cow}{consumer}       {Tr=2000, label={Cow}}
    \esnode{market}{transaction} {Tr=5000, label={Market}}

    \esflow{sun}{grass}{energy}
    \esflow{grass}{cow}{energy}
    \esflow{cow}{market}{energy}
    \esflow{market}{grass}{money_feedback}
\end{energese}
```

which renders as:



That is the same system as `examples/simple_system.json`, and it rasterises identically — the repository's test suite asserts as much.

Macro	Purpose
`\esnode{<id>}{<type>}{<keys>}`	Declare a node.
`\esflow{<from>}{<to>}{<type>}{<keys>}`	Declare an edge.
`\esmeta{<keys>}`	Set diagram metadata.
`\esboundary{<keys>}`	Add a system boundary.

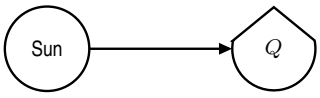
Table 1: The TEX authoring macros.

The trailing key list is optional on every macro. A value containing a comma must be braced — `label_options={midway, above}` — and a bare key is a flag, so `system_boundary` means `system_boundary=true`. Call `\esboundary` more than once for several boundaries.

3.4 Raw TikZ

The symbols and line styles are ordinary TikZ, so a diagram that wants no automatic layout can be drawn by hand:

```
\begin{tikzpicture}
    \node[energese source]  (sun) at (0,0) {Sun};
    \node[energese storage] (q)   at (3,0) {$Q$};
    \draw[energese energy] (sun.energy_out) -- (q.energy_in);
\end{tikzpicture}
```



4 The model

4.1 Nodes

4.2 Edges

Four pathway types are recognised. `energy` and `material` are solid arrows and both participate in the layout. `information` is a solid arrow. `money_feedback` is dashed and, unless the edge routes itself, arcs over the top of the diagram — money runs counter to energy, so it needs to stay clear of it.

Key	Required	Meaning
`id`	yes	Unique identifier, referenced by edges.
`type`	yes	One of the types in section 10.
`Tr`	yes†	Transformity; sets the column.
`label`	yes†	Label text; LaTeX is permitted.
`short_label`	no	Odum's nomenclature for the node: `Q1`, `X2`.
`label_position`	no	`inside`, `above` or `below`.
`seller`, `buyer`	no	On a `transaction`: makes it an exchange.
`energy_label`, `money_label`	no	Captions for an exchange's two sides.
`magnitude`	no	Scale factor for the symbol. Default `1.0`.
`x`, `y`	no	Manual coordinates, overriding the layout.
`label_options`	no	TikZ options applied to the node.
`clearance_w`, `clearance_h`	no	Override the estimated extents used for routing.
`inside`	no	Whether the system window encloses this node. Sources default to `false`, ever
`parent`	no	On an `interaction` or `transaction`: the component to attach to.
`parent_dx`, `parent_dy`	no	The offset from that parent, in diagram units.
`attach`	no	`false` leaves a modifier in a column of its own.

Table 2: Node keys. † not required for `text`, `box` or `ground` nodes, which carry no transformity.

Key	Required	Meaning
`from`, `to`	yes	Node identifiers.
`type`	yes	Pathway type; see below.
`label`	no	Label text.
`label_options`	no	TikZ options for the label node. Supplying this places the label by hand; without
`options`	no	TikZ options for the path, e.g. `bend left=20`.
`volume`	no	Line width in points. Default `1.0`.
`from_anchor`, `to_anchor`	no	Override the chosen anchors.
`waypoints`	no	Explicit geometry; overrides automatic routing.

Table 3: Edge keys.

Every pathway is drawn at `\energeseflowwidth` with the same arrowhead, whether it ends at a symbol or at the ground: they are all flows, and drawing dissipation differently would imply it is something else. Set `volume` on an edge to widen it where the model wants to show relative flow.

Pathways are routed *around* any symbol that is not their own endpoint. In ESL a line meeting a symbol denotes flow into it, so a pathway crossing an unrelated symbol would assert a connection the model does not contain.

An edge that sets its own `bend` is checked too: the package keeps the direction of curvature you asked for and opens the angle until the pathway clears, falling back to automatic routing only if no angle does.

4.3 Exchanges

An exchange is one transaction with two coupled sides: goods pass one way, money the other, and the diamond is where they meet. Declaring the two as separate pathways would say something weaker — that a payment and a delivery both happened, not that they are the same transaction.

Give a `transaction` node a `seller` and a `buyer` and the package draws all four half-pathways:

```
\esnode{sale}{transaction}{Tr=5000, seller=agri, buyer=consumer,
                           money_label={\$}, energy_label={food}}
```

The diamond is a *marker on the pathway*, not a component the pathway enters: goods run from seller to buyer as one continuous line and the diamond rides it at the midpoint, rotated to the flow, with the line

passing through the glyph. Every other symbol is routed around for the good reason that a line meeting a symbol denotes flow into it — but an exchange annotates a coupling rather than transforming anything, so there is nothing to flow into. It accordingly carries no transformity and no heat sink.
producing `seller` → diamond → `buyer` in energy, and `buyer` → diamond → `seller` in money. The legs are routed, checked and drawn like any other pathway, and run tightly parallel with the money just outside the goods, so that the pair reads as one transaction rather than as two pathways that happen to coincide.

4.4 Geometry from the model

Automatic layout is a default, not an obligation. A modeller working in a drag-and-drop editor has already made every placement decision, and a renderer that re-derived them would publish something other than what was drawn. Geometry may therefore be supplied at both levels:

- `x` and `y` on a node override the layout engine, per axis and independently, so one component can be placed by hand while the rest are laid out automatically.
- `waypoints` on an edge override the router. In JSON they are a list of pairs, `"waypoints": [[3, 2], [5, 4]]`; in TEX the same written compactly, `waypoints={3,2; 5,4}`.

The two mix freely. A model carrying complete geometry renders exactly as it was drawn; one carrying none is laid out from scratch.

4.5 Metadata

`column_spacing` and `row_spacing` tune the layout grid; `show_heat_sink` (default true) draws the dissipation pathways and their common ground; `heat_sink_label`, `heat_sink_y` and `ground_x` position and name it; `system_boundary` draws one or more enclosing rectangles, each taking `x_min`, `x_max`, `y_min`, `y_max`, and optionally `label` and `style`.

Setting `system_boundary` to `true` fits the frame automatically: a fixed margin outside the extents of everything the model puts inside the system, with room underneath for the dissipation funnel, which converges inside the frame so that a single line crosses it. Sources are outside by convention — a source is an input from beyond the system — and any node overrides that with `inside`. `boundary_margin` sets the margin in diagram units.

5 Labels

Odum names symbols as well as describing them: storages $Q_1, Q_2, \ldots$, sources and flows X and J. Both belong in the model — the short name is what the equations refer to, the long one is what a reader needs — so a node may carry `label` (the friendly name) and `short_label` (Odum's).

`label_mode` in the metadata chooses between them:

`label_mode`	Inside the symbol	Outside
`long` (default)	the friendly name	only if it will not fit
`short`	Odum's name	nothing
`both`	Odum's name	the friendly name

A symbol grows to hold its label, but only as far as its grid cell; a label larger than that would stop the symbol reading as the cell it occupies, so it is placed outside instead — below by default. That decision is automatic, and a node overrides it with `label_position`.

A displaced label is treated as part of its symbol for routing, so pathways go around the text as well as the glyph.

6 The conventions this package implements

Odum's Energy Systems Language has no written grammar. Its conventions are real but were never stated as rules, so any implementation has to decide what they are. This section records the decisions this package makes, so that a diagram it produces can be read — and disagreed with — precisely.

6.1 Horizontal position is ordinal in transformity

Distinct `Tr` values are sorted and assigned to consecutive columns at equal spacing. Transformities of 1, 100, 2000 and 10 000 become columns one to four.

Placement is therefore neither linear nor logarithmic in transformity. Linear placement is unusable, because transformity spans orders of magnitude and every low-quality component would collapse into the left margin. A logarithmic axis would be legible but would assert a metric the language never defines. Rank encodes exactly what the convention claims — the ordering — and nothing further.

Nodes sharing a transformity share a column, which makes a column an equivalence class of energy quality rather than a coordinate.

6.2 Vertical position is a role, not a quantity

Nothing in ESL assigns a magnitude to height, so the vertical axis carries no quantity of its own. In particular it does not carry power: along a transformation chain transformity rises because available power falls, so power is already determined — inversely — by horizontal position, and encoding it vertically as well would be redundant or contradictory.

The vertical axis is not arbitrary either. It carries the same quality ordering as the horizontal, applied *relative to the main chain*, so the energy hierarchy runs from bottom-left to top-right.

Layer	Rule
On the axis	lies on the longest chain of *energy* pathways
Above	higher transformity than the chain node it connects to
Below	receives from the chain at equal or lower transformity
Fanned	anything the rules cannot classify, in declaration order

"Above" captures high-quality inputs, controllers and higher-quality products; "below" captures dissipation, decomposition and waste storage. This is why Odum draws a high-transformity fuel source entering a diagram from the top, and why the Sun — the largest power input to an ecosystem but the lowest in quality — sits at the bottom of the source column.

The chain is built from `energy` pathways only. A flow of degraded material to a waste store is a side branch, and counting it would let such a store capture the chain and sit on the axis when it belongs below it.

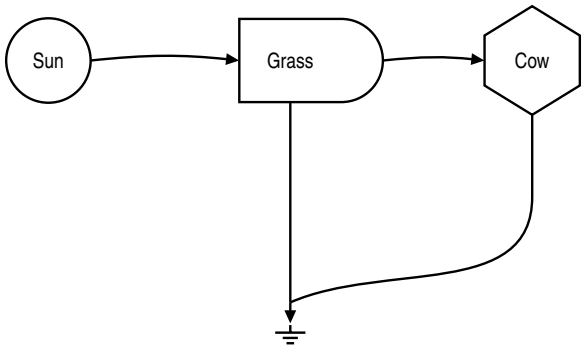
Set `layer_hint` to `control` or `decomposition` to name a layer explicitly; it wins over the derived one, including for a node lying on the main chain. `y_gravity` nudges a node, and an explicit `y` overrides everything.

Fanning uses declaration order, never node identifiers: a layout that changed when a node was renamed would not be reproducible.

6.3 The ground is not a component

If the horizontal axis orders components by energy quality, the heat sink at bottom centre looks like a violation — dissipated energy is the lowest quality present and ought to sit at the far left.

It is not a violation, because the ground is not a component. It carries no transformity, takes no part in the ordering, and terminates pathways rather than transforming anything:



The three components are ordered left to right by transformity. Dissipation runs downward — the vertical convention — and converges on a ground placed by neither rule. Three node types are annotations or terminators rather than components and are exempt from carrying transformity: `ground`, `text` and `box`.

7 Reading a GSSK model

GSSK is the simulation kernel this notation is drawn for, and its model files are read directly:

```
\renderEnergese{model.json}
```

The same macro, because a GSSK file is normalised into an energese model before anything else looks at it. The point is that one file both simulates and draws, so a published figure comes from the model that produced the results beside it and the two cannot drift.

What the conversion does, beyond renaming fields:

- **Presentation is read out of `visual`.** GSSK's schema forbids extra keys on a node, and reserves `visual` for exactly this, so that is where `label`, `Tr`, `magnitude` and the rest belong. The file stays valid for the kernel.
- **Transformity is derived where the model has none**, as the length of the longest chain of pathways reaching each component. That orders the diagram; it is not an emergy calculation, and the values are $1, 2, 3$ rather than plausible-looking figures precisely so that nobody mistakes it for one. Supply `visual.Tr`, or a map under `metadata.energese.transformity`, to say otherwise.
- **Heat sinks are not carried across.** GSSK models dissipation as a `sink` node with pathways into it; this package knows a storage dissipates and draws the pathway and the ground itself. Both would draw it twice, so the sink and its pathways are dropped.
- **Canvas coordinates are converted**, from pixels with y downwards into diagram units with y upwards. The placement you made in an editor is kept; `metadata.energese.layout = auto` discards it and lets the transformity grid lay the diagram out instead, and `metadata.energese.pixels_per_unit` (default 50) sets the scale.

`docs/gssk-interop.md` in the repository has the full field mapping and the parts not yet implemented.

8 Energetic checks

Odum's stated reason for devising ESL was that scholars build models which are energetically invalid. A drawing tool cannot help, because it renders whatever is drawn. This package checks what a renderer can check without leaving the diagram:

1. Energy quality does not fall along an `energy` pathway.
2. Sources take no energy inflow; grounds emit none.

3. Every component is reachable from a source.

4. Energy that arrives somewhere leaves, as outflow or as dissipation.

Findings are reported as warnings, because a diagram that breaks a convention is often still worth drawing and the author is better placed to judge. Set `strict` in the metadata to make them errors instead.

8.1 What is deliberately not checked

`material` pathways are exempt from the first rule. Degraded material moving to a waste or decomposition store genuinely falls in transformity — that is what decomposition is — so applying the rule to material flows reports correct models as broken.

Conservation of energy is not checked. Balancing inflow against outflow and dissipation requires quantities, and a diagram carries none: `volume` sets a line width, not a flux. Conservation belongs with a simulation kernel working from the same model.

Emergy is not checked at all. The algorithms are not standardised and there is no formal definition to validate against, so a check would assert a standard that does not exist. Transformity positions a node; the package neither computes nor validates emergy.

The blunt summary: **if the model is energetically invalid, so is the diagram.** The package renders faithfully and does not repair.

9 Layout

Horizontal position comes from transformity. The distinct `Tr` values in the diagram are sorted and each is assigned a column, so position reflects energy quality rather than the order nodes were declared. Columns are `column_spacing` apart.

Vertical position comes from the central spine method. The longest chain of energy and material pathways is found and laid on the axis; everything else is distributed above and below its column's spine at multiples of `row_spacing`. A node's `layer_hint` (`control` or `decomposition`) nudges it one row up or down, and `y_gravity` shifts it by an arbitrary amount.

Giving a node explicit `x` or `y` overrides the computed value for that axis alone, so a single stubborn component can be placed by hand without abandoning the layout for the rest.

9.1 Modifiers travel with what they modify

An interaction is where two flows combine on their way into the thing they drive, and Odum draws it against that thing; a standalone transaction annotates a pair of parties in the same way. Given columns of their own by transformity, both drift away from the component they modify.

Interactions and transactions are therefore attached: their position is locked to a parent's, and they take no part in the column and row algorithms. The parent is the component they exchange energy with — a consumer for preference — and the offset follows the flow, upstream of the parent if the modifier feeds it and downstream if the parent feeds the modifier. Name a different parent with `parent`, change the offset with `parent_dx` and `parent_dy`, and switch the whole thing off for one node with `attach=false`. An explicit `x` or `y` also wins, as everywhere else.

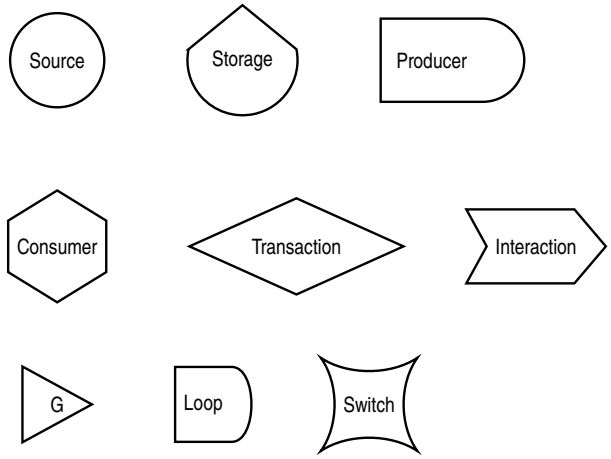
9.2 Pathways and their labels

A pathway that would cross a symbol which is not its own endpoint is routed around it: in this notation a line meeting a symbol means flow entering that symbol, so crossing one asserts a connection the model does not contain. The route is the shortest one that avoids every such symbol. An edge with its own `options` (a `bend`, say) keeps the author's curvature, and the angle is opened until it clears; `waypoints` hands the geometry over entirely.

A pathway's `label` is placed automatically, clear of the symbols, the other pathways, the dissipation bundle and the system frame so far as the diagram allows. Where symbols stand closer together than the

label is wide there may be nowhere clear, and the least obstructed position is used. Supplying `label_options` places the label by hand instead.

10 Symbols, anchors and ports



Every symbol sizes itself to its label, growing its outline around the text so the label always sits inside. `minimum width` and `minimum height` set a floor for symbols with little or no text.

Each flow symbol carries the anchors `energy_in`, `energy_out`, `money_in`, `money_out`, `feedback_in` and `heat_sink`, in addition to `center`, `north`, `south`, `east` and `west`. `feedback_out` is defined on every symbol except `source` and `interaction`. Energy enters on the left and leaves on the right; money moves through the upper half; dissipation leaves through `heat_sink` at the bottom.

10.1 Ports

Every symbol carries a *connection ring*: points spaced a fixed distance apart along its outline, named by index as `ring0` at 0 degrees, counting anticlockwise. The spacing is `\energeseanchorpitch` (3 mm by default), so a larger symbol carries more points than a small one and a coupled pair attaching at adjacent indices is separated by the same distance whichever symbol it meets. The count is rounded to a multiple of four, which puts a point exactly on each axis and holds the spacing to within one point in four of the pitch.

These points are the symbol's *ports*, and they are the only places a pathway attaches. Two consequences follow, and they are what the renderer guarantees:

- **A named anchor is a port.** `energy_in` names a direction — where energy belongs, as Odum draws it — and the pathway attaches at the port nearest that direction. For the anchors on an axis that is the same point; elsewhere it is within half a pitch of it.
- **A port holds one pathway.** The second pathway to want a port takes the next one along, on the side it approaches from, so a fan of flows leaves a symbol as a fan rather than as one line drawn several times.

Neither is something to arrange by hand: `\renderEnergese` and the `energese` environment allocate ports for every pathway they draw. Hand-drawn TikZ can use the same machinery, and address a port directly where the choice of connection point is geometric rather than semantic:

```
\draw[energese energy] (a.ring3) -- (b.ring9);
\draw[energese money_feedback] (b.ring10) -- (a.ring2);
```

or ask for one and let the allocator place it:

```
\energeseresetports                    % begin a diagram
\energeseport{a}{energy_out}{b}        % a's port, facing b
\let\from\energeseportname
\energeseport{b}{energy_in}{a}
\draw[energese energy] (\from) -- (\energeseportname);
```

A symbol whose ports are all taken warns and doubles up rather than dropping the pathway. `box`, `text` and `ground` have no ring, and pathways meet them at the named anchor.

`gain` is the constant-gain amplifier, `loop_limited` the loop-limited converter and `switch` the on/off switching action. Because a triangle is a poor container for text, label a `gain` with a single character as Odum does; a long label there will overflow the apex.

Three further node types are plain TikZ rather than Odum symbols: `box` (a rectangle), `text` (an unboxed label, useful as a routing waypoint) and `ground` (the heat-sink terminator glyph).

Symbol outlines are drawn at `\energeselinewidth`, which defaults to 0.9 pt to match the constant pen weight measured in Odum's published sheet. Redefine it with `\setlength` for a heavier or lighter house style.

Every symbol holds its shape. A shape has a canonical aspect ratio measured from Odum's sheet, and it keeps that ratio at every size: a symbol grows uniformly to hold a longer label rather than stretching, because a widened hexagon reads as a different symbol. Where a label will not fit even so, the label is displaced rather than the shape distorted.

`magnitude` scales the symbol and not its label, so symbols of different magnitude carrying the same words are set in the same type size.

Sizes are set on an invisible grid. Each shape encloses its label differently, so sizing every symbol purely from its own text would make identical labels produce very different symbols. `\energeseunit` is one grid cell (default 1.1 cm), and each symbol's minimum size is the multiple of a cell at which it lands on the relative height Odum draws it at. A label that will not fit its cell grows its own symbol. To scale a whole diagram's symbols together, set the unit:

```
\setlength{\energeseunit}{1.4cm}
```

Note that TikZ's `font` key is a single value, so a `label_options` of `font=\large` replaces the style's `font=\sffamily` rather than adding to it. Write `font=\sffamily\large` to keep the sans face.

11 Heat sinks and the system window

Unless `show_heat_sink` is false, every dissipating component runs a pathway downwards which converges, with all the others, on a single ground symbol. Dissipation is flow to a common sink, not contact with a surface, so there is no horizontal floor line.

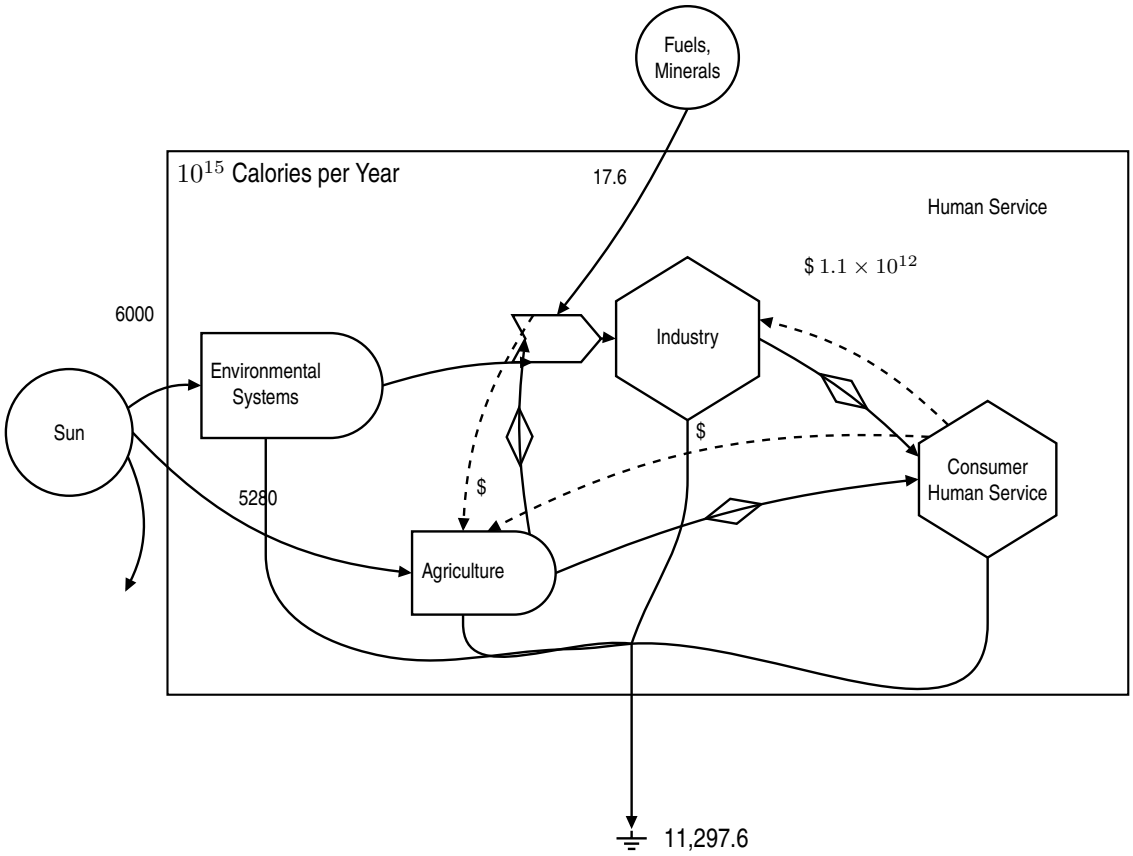
The dissipating types are `producer`, `consumer`, `storage`, `transaction`, `gain`, `loop_limited` and `switch`. Sources and interactions dissipate nothing in Odum's convention, and `text`, `box` and `ground` are plain shapes with no `heat_sink` anchor.

Where a system window is drawn, the dissipation pathways converge just inside it and a single line crosses the frame to reach the ground outside. Each pathway crossing separately would shred the boundary, and would misstate the accounting: what leaves the system is one aggregate flow of used energy, not several.

`system_boundary` draws an optional *system window*: the frame declaring what the model treats as inside the system, and therefore what its macro-level inputs and outputs are. The ground symbol is always placed *outside* that window — used energy leaves the system, so its terminator cannot sit within the frame. A `heat_sink_y` that would put the ground inside the window is overridden rather than honoured.

12 A worked example

`examples/aggregated-economy.json` reproduces a classic Odum figure. It exercises manual coordinates, a system boundary, a relocated heat sink and several money-feedback arcs:


Fuels,
Minerals
10^{15} Calories per Year
17.6
Human Service
$\$ 1.1 \times 10^{12}$
6000
Industry
Environmental
Systems
Sun
$
Consumer
Human Service
5280
$
Agriculture
11,297.6